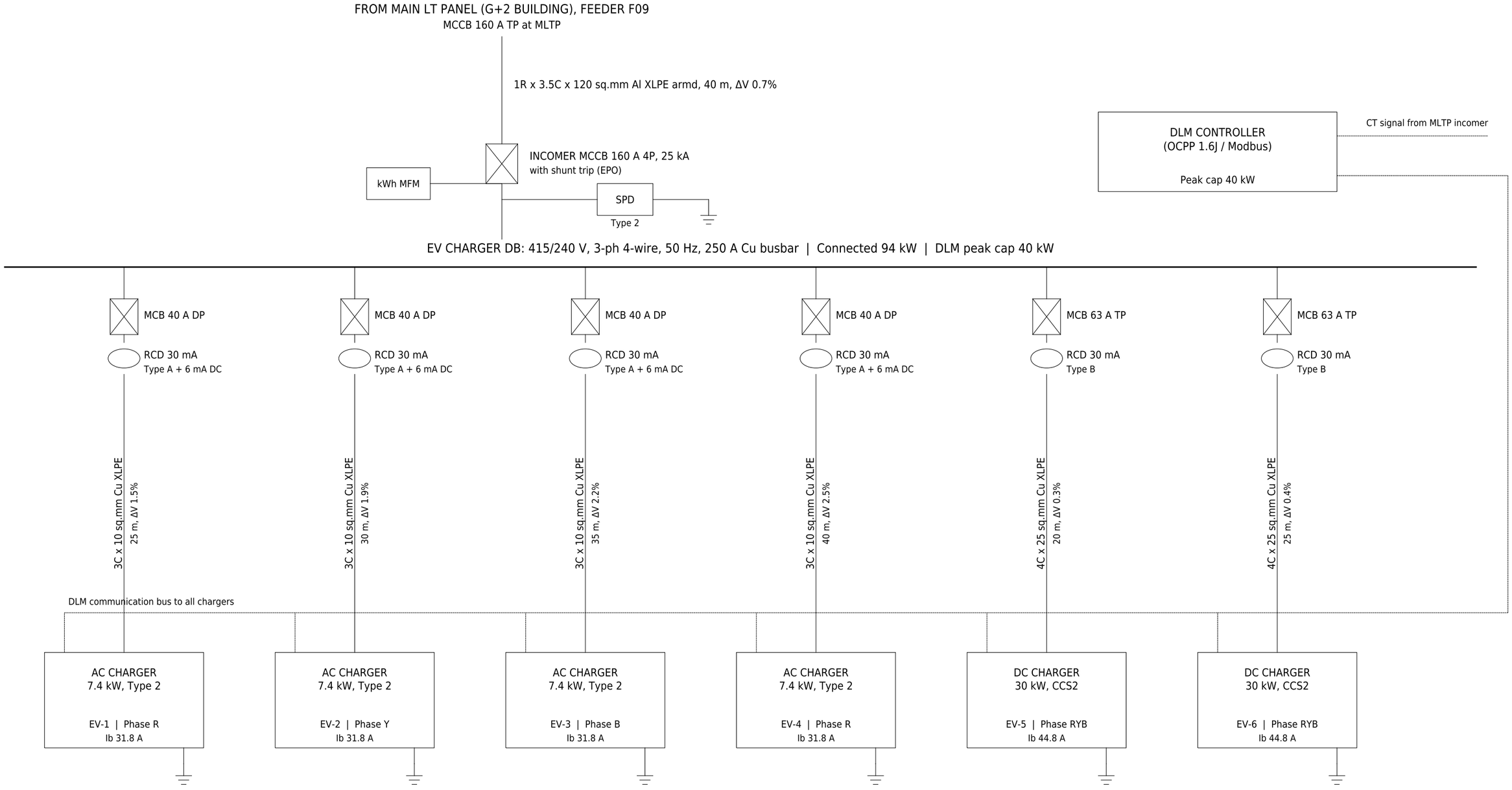
SINGLE LINE DIAGRAM: EV CHARGER DISTRIBUTION BOARD (6 CHARGING POINTS)



NOTES

1. Design and ratings from EV_Charging_Design_Calcs.xlsx. Host building: G+2 commercial office (see companion project), feeder F09.
2. Without load management the 94 kW of chargers would require a 400 kVA transformer; DLM caps EV at 40 kW at office peak so the 315 kVA unit is retained.
3. Feeder sized on full connected load (no DLM credit). One dedicated circuit, RCD and protective earth per charger (IEC 60364-7-722).
4. Final circuits: Cu XLPE armoured; In ≥ 1.25 x Ib (continuous load); ΔV ≤ 3%. EV DB is non-essential: not on DG.
5. Two AC chargers share phase R; DLM allocates current phase-aware. Emergency power-off trips incomer via shunt trip.

PROJECT	EV Charging, G+2 Office, Delhi
DRAWING	SLD, EV Charger DB
DRAWN BY	Anuj Kumar
SCALE	NTS SHEET: A3
DWG NO.	EV-EL-SLD-001 REV 0